Task 5 — Reliability Sign-off & Scale Integration

Data Analyst · Sprint A - Scale & Reliability · Task Brief

DATA ANALYST PlaceMux · Phase 3 · Task Brief

TASK 5 / 25 **Advanced · Advanced** Sprint A - Scale & Reliability · Mastery 67%

Focus

Sign off data reliability: the numbers used for scale decisions are complete, fresh and trustworthy.

What to build / complete

- Verified data completeness and freshness across critical pipelines
- Reconciliation between source systems and the warehouse
- A data-reliability sign-off

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: Every decision-critical number is proven complete and current - or explicitly flagged as not.

Stage B — Build: Verified data completeness and freshness across critical pipelines

1. Define the metric, its source & the decision

Define precisely what “Verified data completeness and freshness across critical pipelines” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: Reconciliation between source systems and the warehouse

1. Define the metric, its source & the decision

Define precisely what “Reconciliation between source systems and the warehouse” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: A data-reliability sign-off

1. Define the metric, its source & the decision

Define precisely what “A data-reliability sign-off” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Reconcile warehouse counts against the source system live and show them matching.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “Verified data completeness and freshness across critical pipelines” is complete, real, and demoable end-to-end.
- “Reconciliation between source systems and the warehouse” is complete, real, and demoable end-to-end.
- “A data-reliability sign-off” is complete, real, and demoable end-to-end.
- Good looks like: Every decision-critical number is proven complete and current - or explicitly flagged as not.

Worry if any of these are true

- Warehouse figures not reconciled to source systems.
- Known data gaps not surfaced on dashboards.
- Calling an experiment early because it looked good on day two.

Hand-off

- Trusted data baseline to all roles. Make sure the next person can pick it up without you.